

PLANT PHYSIOLOGY

TRANSPORT IN PLANTS

1. Over small distance, substances can move by :
 - (1) Diffusion
 - (2) Cytoplasmic streaming
 - (3) Active transport
 - (4) All of the above
2. Substances that have a..... moiety, find it difficult to pass through the membrane.
 - (1) Hydrophilic (2) Hydrophobic
 - (3) Neutral (4) Lipophilic
3. Transport methods those require special membrane proteins also show :
 - (1) Always uphill movement
 - (2) Always movement according to concentration gradient
 - (3) Always transport saturation
 - (4) Always ATP expenditure
4. Which of the following statements are correct?
 - (A) If two systems containing water are in contact, random movement of water molecules will result in net movement of water down a gradient of free energy is called diffusion.
 - (B) The less the solute molecules in a solution, the lower is the solute potential
 - (C) If a pressure greater than atmospheric pressure is applied to a solution, its water potential increases.
 - (D) By convention, the water potential of pure water at standard temperature which is not under any pressure, is taken to be zero i.e. minimum value of water potential.
 - (1) A and B (2) A and C
 - (3) B and D (4) C and D
5. Imbibition :
 - (1) is a special type of osmosis
 - (2) involve adsorption
 - (3) is the characteristic feature of lipophilic colloids
 - (4) occurs against the water potential gradient
6. All the following statements are correct except that :
 - (1) the symplastic movement of absorbed water may be aided by cytoplasmic streaming
 - (2) the xylem vessels and tracheids are non living conduits so are parts of the apoplast
 - (3) the movement through the apoplast does not involve crossing the cell membrane
 - (4) the apoplastic system is the system of interconnected protoplasts that is continuous through the plant, except at the casparian strips of the endodermis in the roots.
7. As various ions from the soil are actively transported into the vascular tissues of the roots, water follows and increases the pressure inside the xylem. This pressure :
 - (1) is responsible for water loss from leaves in liquid phase
 - (2) may re-establish the continuity of water column in xylem
 - (3) is considered as positive pressure
 - (4) All of the above
8. The cause of the opening or closing of the stomata is:
 - (1) a change in the turgidity of the guard cells
 - (2) the crescent shape of thick and nonelastic outer wall of the guard cells
 - (3) the longitudinal orientation of cellulose microfibrils in the inner walls of guard cells
 - (4) All of the above
9. Most of the nitrogen in plants is transported in :
 - (1) Organic form via phloem
 - (2) Organic form via xylem
 - (3) Inorganic form via phloem
 - (4) Inorganic form via xylem
10. In plants the accepted mechanism for the translocation of sugars from source to sink :
 - (1) involves the modest push by root pressure
 - (2) involves the transport according to pressure potential gradient
 - (3) is completely based upon transpiration pull
 - (4) Does not requires metabolic energy

- 11.** Mineral translocation in plants is carried out by :
- Xylem exclusively
 - Phloem exclusively
 - Mainly xylem & little bit by phloem
 - Mainly phloem & little bit by xylem
- 12.** Diffusion is very important to plants since it is the only means for :
- Water translocation in root
 - Gaseous movement within plant
 - Mineral translocation in root
 - Sugar transport from source to sink
- 13.** Which of the following is not a similarity between facilitated diffusion and active transport ?
- Transport saturation
 - Sensitivity towards protein inhibitors
 - Selectivity
 - Uphill transport
- 14.** Water will move from it's region of :
- lower ψ_p to higher ψ_p
 - lower ψ_s to higher ψ_s
 - lower ψ_w to higher ψ_w
 - higher ψ_w to lower ψ_w
- 15.** Which of the following is ultimately responsible for enlargement of plant cells ?
- Osmotic pressure
 - Turgor pressure
 - Wall pressure
 - Osmotic potential
- 16.** Beside water potential gradient, which of the following is also prerequisite for imbibition ?
- permeable membrane
 - impermeable membrane
 - affinity between adsorbant & liquid
 - selectively permeable membrane
- 17.** Regarding mycorrhiza select out the incorrect statement :
- they have large surface area
 - the fungus provides minerals & water
 - roots provide nitrogenous compounds
 - it can never be of obligate nature
- 18.** Which of the following is not observed during stomatal opening ?
- High turgidity of guard cells
 - Radially oriented microfibrils
 - Outer wall bulge out
 - Low turgor of guard cells
- 19.** Which of the following is not a significance of transpiration ?
- Absorption of water
 - Absorption of minerals
 - Cooling of leaf surface
 - Maintain the shape and structure of plant

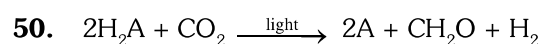
MINERAL NUTRITION

- 20.** Which of the following is the method by which essential elements were identified in plants ?
- Plant ash analysis
 - Hydroponics
 - Plant tissue culture
 - Nitrogen fixation
- 21.** Which of the following essential elements is required by plants in excess of $10 \text{ m mole kg}^{-1}$ of dry matter?
- Magnesium
 - Manganese
 - Molybdenum
 - Selenium
- 22.** Choose the pair from the following in which one element is essential to plant while other is beneficial but not essential.
- Copper and Molybdenum
 - Sodium and Silicon
 - Chlorine and Cobalt
 - Selenium and Cobalt
- 23.** Which of the following element is an activator for both ribulose biphosphate carboxylase oxygenase enzyme and phosphoenol pyruvate carboxylase enzyme?
- Zinc
 - Copper
 - Magnesium
 - Chlorine
- 24.** Choose the pair from the following in which both the elements share common function during photosynthesis in plants.
- Chlorine and Magnesium
 - Potassium and Phosphorus
 - Boron and Molybdenum
 - Manganese and Chlorine

- 25.** If deficiency symptoms of any element appear first in the senescent leaves, this element should not be :
- (1) Calcium (2) Nitrogen
(3) Potassium (4) Magnesium
- 26.** Which of the following bacteria oxidise the ammonia into nitrite?
- (1) *Nitrococcus*
(2) *Nitrobacter*
(3) Both (1) and (2)
(4) *Thiobacillus*
- 27.** Which of the following is correct regarding non leguminous plant *Alnus*?
- (1) Free living nitrogen fixation by *Beijernickia*
(2) Free living nitrogen fixation by *Frankia*
(3) Symbiotic nitrogen fixation by filamentous microbe
(4) Symbiotic nitrogen fixation by *Rhizobium*
- 28.** During biological nitrogen fixation the energy input is :
- (1) 16 ATP for each NH_3
(2) 8 ATP for two NH_3
(3) 32 ATP for two NH_3
(4) 8 ATP for each NH_3
- 29.** During nitrogen metabolism in plants, transaminase enzyme is used in conversion of :
- (1) Glutamic acid into other amino acids
(2) α -Ketoglutaric acid into glutamic acid
(3) Glutamic acid into glutamine
(4) NH_4^+ into glutamic acid
- 30.** Proper aeration is required in hydroponics.
- (1) to avoid the toxicity of minerals
(2) for translocation of mineral from root to shoot
(3) for absorption of minerals
(4) to decrease the osmotic pressure in root cells
- 31.** Which of the following is not a beneficial element for plant life ?
- (1) Na (2) Sr
(3) Si (4) Co
- 32.** Water potential of a cell is mainly determined by which of the following element ?
- (1) Mg^{++} (2) Ca^{++}
(3) K^+ (4) Fe^{2+}
- 33.** The element is said to be deficient, when present:
- (1) below critical concentration
(2) above critical concentration
(3) at critical concentration
(4) both below and above critical concentration
- 34.** Deficiency symptoms of element can be visualised by what kind of changes ?
- (1) Physiological changes
(2) Morphological changes
(3) Chemical changes
(4) Anatomical changes
- 35.** Deficiency symptoms for which of the following element tend to appear first in young tissues?
- (1) N & P (2) N & Ca
(3) Ca (4) S & K
- 36.** Mn toxicity leads to Ca deficiency by :
- (1) competing with Ca uptake
(2) inhibiting translocation to shoot apex
(3) competitive inhibition for enzymes
(4) All of the above
- 37.** What is the major fate of NH_3 produced by ammonification ?
- (1) Volatilise to re-enter in the atmosphere
(2) Absorbed by plants
(3) Converted into nitrates
(4) Denitrification
- 38.** In which of the following root tissues nodule formation is initiated after successful infection ?
- (1) Epidermis (2) Cortex
(3) Endodermis (4) Root hairs
- 39.** First stable product of biological nitrogen fixation is:
- (1) $\text{HN} = \text{NH}$
(2) $\text{H}_2\text{N} - \text{NH}_2$
(3) NH_3
(4) NO_3^- or NO_2^-

PHOTOSYNTHESIS

- 40.** During photosynthesis, plants mainly utilise the red and blue regions of visible spectrum, for the first time it was concluded by :
- (1) Jan Ingenhousz (2) Joseph Priestley
(3) T.W. Engelmann (4) Cornelius Van Niel
- 41.** Which of the following conclusions regarding photosynthesis was proved by using radioisotopic techniques ?
- (1) Light is essential
(2) O_2 comes from H_2O and not from CO_2
(3) Glucose is stored as starch
(4) Exchange of gases with environment
- 42.** The dark reactions of the photosynthesis :
- (1) occur in darkness
(2) are not light dependent
(3) are not directly light driven
(4) occur in membrane system of chloroplast
- 43.** Electrons from which of following reduces $NADP^+$ to $NADPH+H^+$ during Z-scheme of photosynthesis?
- (1) Photosystem-I (2) Water
(3) Carbon dioxide (4) Photosystem-II
- 44.** During photosynthesis the stroma lamellae of chloroplast could perform :
- (1) the process of dark reaction in which ATP utilised
(2) the process of light reaction which produce $NADPH+H^+$
(3) the process of dark reaction which utilise $NADPH+H^+$
(4) the process of light reaction which produce ATP
- 45.** Which of the following statements are correct regarding synthesis of ATP in chloroplast during photosynthesis ?
- (A) Splitting of water in stroma helps in creation of proton gradient
(B) Cytochrome complex helps in the release of protons in the lumen of thylakoid by accepting electrons from hydrogen carrier.
(C) Movement of protons across the membrane to the stroma through the F_0 of the ATPase is coupled with ATP synthesis.
- (D) Reduction of $NADP^+$ to $NADPH+H^+$ is also a cause for creation of proton gradient.
- (1) All statements are correct
(2) C and D
(3) A and B
(4) B, C and D
- 46.** What is the correct ratio of ATP utilisation in steps of Calvin cycle?
- (1) Reduction : Regeneration :: 1 : 1
(2) Reduction : Regeneration :: 2 : 1
(3) Reduction : Regeneration :: 2 : 2
(4) Reduction : Regeneration :: 1 : 2
- 47.** The cells of C_4 plants those are rich in RuBisCO enzyme, also have which of the following characteristic (s)?
- (1) Intercellular spaces absent
(2) Thick walls impervious to gaseous exchange
(3) Large number of chloroplast
(4) All of the above
- 48.** The productivity is better in C_4 plants because :
- (1) they increase the intracellular concentration of CO_2 in mesophyll cells
(2) in these plants RuBisCO has much greater affinity for O_2 than for CO_2
(3) these plants can prevent competitive binding phenomena related to RuBisCO
(4) these plants minimise the carboxylase activity of RuBisCO
- 49.** C_3 plants respond to higher CO_2 concentration by showing increased rates of photosynthesis because:
- (1) Current availability of CO_2 levels is limiting to the C_3 plants
(2) C_3 plants show saturation at about $360 \mu/L^{-1}$ concentration of CO_2
(3) these plants responds to high CO_2 concentration even in low light conditions
(4) in these plants RuBisCO shows only carboxylation



in this given equation H_2A represents to :

- (1) Suitable reducible compounds
 - (2) Suitable oxidisable compound
 - (3) Suitable buffer
 - (4) Both (1) and (2)
51. The membrane system of chloroplast is responsible for :
- (1) Trapping the light energy
 - (2) Synthesis of ATP & NADPH
 - (3) Enzymatic reactions for CO_2 incorporation
 - (4) Both (1) and (2)
52. How does PS-II supply electrons continuously ?
- (1) by removing electrons from photon
 - (2) by removing electrons from H_2O
 - (3) by removing electrons from CO_2
 - (4) by removing electrons from constituent carotenoids
53. Which of the following is not always required for chemiosmosis ?
- (1) Membrane
 - (2) Proton pump
 - (3) OEC
 - (4) ATPase
54. Classification of biosynthetic phase of dark reaction as C_3 & C_4 is primarily based on.
- (1) Initial CO_2 fixation
 - (2) Final CO_2 assimilation
 - (3) First CO_2 receptor
 - (4) Number of ATP get consumed
55. Which of the following is not special about C_4 plants?
- (1) Responsiveness to high light intensities
 - (2) Lack of photorespiration
 - (3) Greater productivity
 - (4) Scotoactive stomata

56. In C_4 plants there is no photorespiration, because :

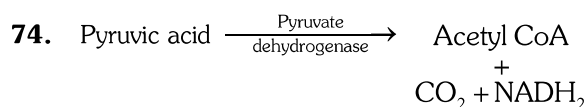
- (1) They have large number of chloroplast
 - (2) Increased CO_2 concentration at RuBisCO site
 - (3) Concentric arrangement of mesophyll cells
 - (4) Greater affinity of RuBisCO for CO_2
57. Which of the following is not a plant factor regulating photosynthesis ?
- (1) Age of leaf
 - (2) Number of mesophyll cells
 - (3) Atmospheric CO_2 concentration
 - (4) Amount of chlorophyll
58. Increase in CO_2 concentration upto percent can cause an increase in CO_2 fixation rate, beyond this the level can become damaging over long periods.
- (1) 0.03 percent
 - (2) 0.04 percent
 - (3) 0.045 percent
 - (4) 0.05 percent

RESPIRATION IN PLANTS

59. The energy released by oxidation of respiratory substrates :
- (A) Comes out in a single step to increase the possibility of maximum ATP production
 - (B) is not used directly
 - (C) is used directly in the energy requiring processes of the organisms
 - (D) is trapped as chemical energy in the energy currency of the cell
- (1) C and D are incorrect
 - (2) B and D are correct
 - (3) A and B are correct
 - (4) A and D are incorrect
60. How many ATP molecules and during which steps, are directly synthesised in EMP pathway from one glucose molecule?
- (1) 4 ATP, 2 in each PEP to pyruvic acid and BiPGA to PGA
 - (2) 8 ATP, 4 in each PEP to pyruvic acid and BiPGA to PGA
 - (3) 2 ATP, 1 in each Glucose to Glucose-6-P and Fructose-6-P to Fructose 1, 6 BiP
 - (4) 4 ATP, 2 in each Glucose to Glucose-6-P and Fructose-6-P to Fructose 1, 6 BiP

- 61.** Which of the following enzyme(s) is/are involved in the conversion of pyruvic acid into CO_2 and ethanol?
- Pyruvic acid dehydrogenase
 - Alcohol decarboxylase
 - Both (1) and (2)
 - Pyruvic acid decarboxylase
- 62.** The complete oxidation of one molecule of pyruvate by the stepwise removal of all the hydrogen atoms:
- leaving six molecules of CO_2
 - leaving two molecules of CO_2
 - leaving four molecules of CO_2
 - leaving three molecules of CO_2
- 63.** In aerobic respiration, the ultimate or final electron acceptor is :
- Atomic oxygen
 - Molecular oxygen
 - Cytochrome a_3
 - Water
- 64.** Fermentation differs from aerobic respiration :
- in having partial breakdown of glucose
 - in producing less ATP per glucose
 - in having slow oxidation of NADH_2 to NAD^+
 - All of the above
- 65.** Complete oxidation of which of the following respiratory substrate evolve less volume of CO_2 as compare to volume of O_2 consumed ?
- Fats
 - Proteins
 - Carbohydrates
 - Both (1) and (2)
- 66.** Enzymes differ from inorganic catalysts because enzymes get damaged at high temperatures. This difference :
- is applicable to all enzymes
 - is not applicable to thermolabile enzymes
 - is not applicable to the enzymes of thermophilic organisms
 - is applicable to thermostable enzymes
- 67.** An enzyme catalysing a transfer of a group, G between a pair of substrate S and S' as follows :
- $$\text{S} - \text{G} + \text{S}' \longrightarrow \text{S} + \text{S}' - \text{G}$$
- G = phosphate or hydrogen or any other group, the enzyme is related with which of the following class?
- Transferases
 - Dehydrogenases
 - Both (1) and (2)
 - Either (1) or (2)
- 68.** Enzyme, which catalyzes the breakdown of hydrogen peroxide to water and oxygen, has which type of cofactor?
- Tightly bound inorganic compound
 - Tightly bound organic compound
 - Permanently bound inorganic compound
 - Loosely bound organic compound
- 69.** What is the significance of respiration ?
- Production of cellular energy currency
 - Provides carbon skeleton as precursor for synthesis of various chemicals
 - loss of weight
 - Both (1) and (2)
- 70.** Plants donot present great demands for gaseous exchange because :
- They are autotrophic
 - Photosynthesis and respiration work mutually
 - In plants there is less need of energy
 - Plants are regulators
- 71.** Select out the correct sequence of glycolytic steps:
- $\text{PGAL} \rightarrow 3\text{-PGA} \rightarrow 1,3\text{-BiPGA} \rightarrow \text{PEP}$
 - $\text{PGAL} \rightarrow 1,3\text{-BiPGA} \rightarrow \text{PEP} \rightarrow 3\text{-PGA}$
 - $\text{PGAL} \rightarrow 1,3\text{-BiPGA} \rightarrow 3\text{-PGA} \rightarrow \text{PEP}$
 - $\text{PGAL} \rightarrow \text{PEP} \rightarrow 1,3\text{-BiPGA} \rightarrow 2\text{-PGA}$
- 72.** During respiration of *Yeast* which of the following enzyme is not used in oxygen stressed conditions ?
- Enolase
 - Pyruvic acid decarboxylase
 - Alcohol dehydrogenase
 - Aconitase

- (1) 7 percent
- (2) less than seven percent
- (3) more than seven percent
- (4) always 2 percent



(1) Mg^{++} (2) NAD^+
(3) Co-A (4) TPP

(1) OAA (2) Water
(3) NAD (4) both (1) and (2)

(1) Oxalosuccinic acid (2) α -ketoglutaric acid
(3) Succinyl Co-A (4) Cis aconitic acid

(1) complex-I (2) complex-II
(3) complex-III (4) complex-IV

(1) 1.0 (2) 0.7
(3) 0.9 (4) 1.3

GROWTH & DEVELOPMENT

(A) In plants, the form of growth is open and localised

(B) Swelling of piece of wood in water is considered as growth since it involve the increase in size

(C) Growth is accompanied by metabolic processes

(D) Growth, at a cellular level, is a result of increase in the amount of protoplasm

- (1) All the statements are correct
- (2) A and B
- (3) B, C and D
- (4) A, C and D

- (1) Increased vacuolation
- (2) Maximal size in terms of protoplasmic modifications
- (3) Cells those are rich in protoplasm and having thin cell walls with abundant plasmodesmata
- (4) Cell enlargement

(A) Sigmoid curve

(B) expressed as $W_1 = W_0 + rt$

(C) Linear curve

(D) Three phases - Lag, exponential and stationary

(1) A and D (2) A, B and D

(3) Only C (4) B and C

- (1) Open indeterminate growth
- (2) Open determinate growth
- (3) Open differentiation
- (4) Capacity of dedifferentiation

(A) Auxin (i) Derivatives of carotenoids
(B) Gibberellin (ii) Gas
(C) Cytokinin (iii) Adenine derivatives
(D) Ethylene (iv) Terpenes
(E) Absciscic acid (v) Indole compounds

(1) A - i, B - ii, C - iii, D - iv, E - v
(2) A - ii, B - i, C - v, D - iii, E - iv
(3) A - v, B - iv, C - iii, D - ii, E - i
(4) A - iv, B - iii, C - i, D - ii, E - v

84. Match the following :

- | | |
|---------------------------------|----------------------|
| (A) Human urine | (i) Ethylene |
| (B) Coconut milk | (ii) GA ₃ |
| (C) Ripened oranges | (iii) Auxin |
| (D) <i>Gibberella fujikuroi</i> | (iv) Cytokinin |

- (1) A - iii, B - iv, C - i, D - ii
 (2) A - i, B - ii, C - iii, D - iv
 (3) A - iv, B - iii, C - ii, D - i
 (4) A - ii, B - i, C - iv, D - iii

85. Match the following :

- | | |
|--------------------|--------------------------------|
| (A) Cytokinin | (i) Weed free lawns |
| (B) Auxin | (ii) Brewing industry |
| (C) Abscissic acid | (iii) Root hair formation |
| (D) Ethylene | (iv) Overcome apical dominance |
| (E) Gibberellin | (v) Stress hormone |

- (1) A - iv, B - i, C - v, D - ii, E - iii
 (2) A - i, B - ii, C - v, D - iii, E - iv
 (3) A - iii, B - iv, C - v, D - i, E - ii
 (4) A - iv, B - i, C - v, D - iii, E - ii

86. Which of the following occur naturally in plants?

- (1) 2, 4-dichlorophenoxyacetic acid
 (2) Indole butyric acid
 (3) Naphthalene acetic acid
 (4) Kinetin

87. Plants, in which there is no correlation between exposure to light duration and induction of flowering response, are called :

- (1) Day - neutral plants
 (2) Long day plants
 (3) Short day plants
 (4) Monocarpic plants

88. Which of the following is not a biennial plant?

- (1) Barley (2) Sugarbeet
 (3) Cabbages (4) Carrots

89. Swelling of piece of wood when placed in water is not considered as growth because :

- (1) It does not occur at expense of energy
 (2) It is not a metabolic change
 (3) It is reversible process
 (4) It is extrinsic process

90. Plant growth is unique because :

- (1) It is intrinsic
 (2) It occurs at the expense of energy
 (3) Plant retains the capacity for unlimited growth throughout their life
 (4) Its accompanied by metabolic processes

91. Growth at a cellular level, is principally a consequence of increase in amount of :

- (1) Cell wall material (2) Water
 (3) Protoplasm (4) Cell sap

92. Cells with increased vacuolation, cell enlargement and new cell wall deposition are the characteristic features of which phase of growth ?

- (1) Meristematic phase
 (2) Elongation phase
 (3) Maturation phase
 (4) Differentiation phase

93. Trees showing seasonal activities, represent what kind of growth curve ?

- (1) Sigmoid (2) Linear
 (3) J-shaped (4) double sigmoid

94. Quantitative comparisons between the growth of living systems can be made by :

- (1) Absolute growth rate
 (2) Relative growth rate
 (3) Exponential growth rate
 (4) Both (1) and (2)

95. Nutrients both macro and micro essential elements are required by plant during growth for :

- (1) Synthesis of protoplasm
 (2) As source of energy
 (3) Enzyme of activation
 (4) All of the above

96. Parenchyma cells that are made to divide under controlled laboratory conditions during plant tissue culture, represents :

- (1) Differentiation
 (2) Dedifferentiation
 (3) Redifferentiation
 (4) Undifferentiated mass of cells

97. Which of the following is an intercellular intrinsic factor regulating development ?

- (1) genetic constitution
- (2) PGR
- (3) Water
- (4) Oxygen

98. Select out the incorrect match :

- (1) GA - speed up malting process
- (2) Auxin - Xylem differentiation
- (3) Cytokinin - Adventitious shoot formation
- (4) Ethylene - Lateral shoot growth